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Failure analysis

Every claim here is traceable to a run under `experiments/`. Numbers are quoted with the dataset they were measured on, because accuracy on this task is meaningless without saying which generator produced the data.

The four measurement domains

The localizer is measured on four datasets built by generators that share no image formation code. This matters: a localizer tuned on one generator will silently learn that generator's physics, and the hidden test set comes from a different one.

Dataset	Generator	What it tests
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`train40_v2`	primary physics generator	modern node dimensions, secant law edge brightening, dose driven noise, scan artifacts
`amat40`	faithful reproduction of the reference starter pipeline	coarse structure presets, area averaged decimation, four documented noise tiers
`spec40`	independent reading of the organiser specification	9 to 1 through 11 to 1 magnification, rotation, the full published degradation list
`stress30`	adversarial generator	painted edges, plain Gaussian noise, harsher pose ranges

Root cause one, the centre tie break overriding a decisive maximum

This was the largest single defect found, and it was found by a test rather than by an accuracy number.

The problem statement requires that when several valid matches exist, the one closest to the search image centre is returned. The implementation defined "several matches" as every local maximum within 0.015 of the best score. On noise free, exactly decimated image pairs whose reference window straddles unique boundaries, so that the correct answer is unique by construction, that window admitted **68 to 81 candidates**, and the tie break then discarded a correlation maximum that was exactly at the true location. Measured errors of 368 px and 102 px on cases whose truth was the global maximum with a score gap of 0.0000.

The reason this is a defect and not a faithful reading of the rule: the reference pipeline samples the reference crop origin uniformly across the fine canvas, so the truth is uniformly distributed in the search frame and carries no bias toward the centre. Proximity to the centre therefore provides no information about correctness, and applying it to candidates that are measurably worse can only lose accuracy. The rule is meaningful only among matches that are genuinely indistinguishable.

The fix restricts the equal match set to numerical ties, so the tie break fires only when the correlation genuinely cannot separate candidates, while the wider noise scaled pool continues to feed the residual disambiguation stage. `experiments/\*\_tolerance\_and\_template\_ablation` records the sweep that selected the threshold.

Root cause two, template formation not matching search formation

The reference pipeline forms the search image by blurring the fine canvas and decimating it with area averaging, so each search pixel integrates a 10 by 10 block of specimen. The template was formed by bilinear point sampling at 10 times decimation, which reads one pixel in a hundred and retains aliasing the search image does not contain. The fix folds the decimation box filter into the template blur in quadrature with the beam spot, so the template carries matched anti aliasing. This is the same class of error as matching a sharp template against a soft image: the correlation peak survives but loses margin, and lost margin is exactly what turns an identifiable site into a near tie.

Root cause three, an over wide pose search

Widening the magnification search to the stated 9 to 1 through 11 to 1 range was necessary, because the previous plus or minus 4 percent would have failed outright on any pair beyond 10.4 to 1. But a wider grid also gives many more chances for a wrong pose to score highly at a wrong location. On a proxy pair whose true magnification was exactly 10 to 1, the search selected a scale of 0.885 and locked onto a site 273 px away.

The fix evaluates the nominal pose first at full resolution and requires an off nominal pose to beat it by a margin before it is accepted. Coverage is retained for genuinely rotated or mis scaled inputs, while the common exact decimation case is both faster and less likely to be captured by a spurious hypothesis. The pose actually used is reported per pair as `pose\_source`.

Root cause four, impulse noise

Salt and pepper noise up to a few percent of pixels appears in the published degradation list. Impulse pixels are unbounded outliers, so they dominate the sums inside normalized cross correlation, and Gaussian denoising spreads them rather than removing them. An adaptive 3 by 3 median that fires only when the impulse fraction is detectable recovered one affected pair from 765 px to 0.09 px.

The ambiguity dominated failure regime

The remaining failures are concentrated in defect free periodic interiors, and three independent tests converge on the same conclusion: they are dominated by information limited spatial ambiguity rather than search range or candidate budget failure.

First, the candidate budget test: raising the number of hypotheses promoted to full resolution from 6 to 12 and 24 produced answers identical to three decimal places on all 150 pairs across four domains, so the correct candidate is not being discarded during prescreening. Second, pose inspection: on the failing 9 to 1 magnification cases the estimated scale lands within 0.007 to 0.017 of truth with the rotation correct, and the prediction still sits on a wrong periodic instance, so the transformation is found and the instance selection is what fails. Third, the oracle experiment below, where supplying the exact true pose from the generator metadata does not eliminate the failures.

In the strictly identical limit the statement is exact: if two candidate windows are the same array of pixels, no function of the images can prefer one. In the realistic near identical regime the supported claim is the measured one: within the available image evidence the competing sites are indistinguishable, and neither a larger search budget nor the true pose resolves them. Localization there is ill posed rather than hard, and the information that disambiguates has to come from aperiodic content: array boundaries, termination structures, missing or oversized contacts, line edge roughness that happens to be locally distinctive.

An oracle experiment quantified this. Supplying the true scale and rotation from the generator metadata, so that the pose search cannot be blamed, and then measuring the correlation at the true location against the global maximum, the failures on the specification proxy split into three classes:

Class	Share	Diagnosis
Search failure	2 of 10	the oracle pose lands within 2 px, so the pose search missed a recoverable answer
Appearance corruption	3 of 10	correlation at the truth is 0.21 to 0.54 even with the oracle pose, all in charging streak or extreme contrast variants
Genuine near tie	5 of 10	correlation at the truth is within 0.02 to 0.05 of a wrong lattice peak

Only the first class is straightforwardly fixable. The third class is the ill posed regime, and the honest engineering response is not to guess better but to report the situation, which the localizer does: every answer carries a regime label of `unique\_peak`, `residual\_identified` or `tie\_break\_convention`. On a real tool that distinction is the difference between trusting a measurement and re acquiring at lower magnification.

What the residual disambiguation stage buys, and where it stops

Where the window does contain distinguishing content, cell to cell reference subtraction recovers it. The shared periodic content is estimated by a pixelwise median over aligned candidate windows and projected out of the template along with its sub pixel shift terms; the remaining deviation field is scored densely at every position in closed form; a robust z score decides whether one candidate stands out.

On the primary dataset this converts lattice mislocks into sub pixel answers and raised the pass rate within 1 px from 82.5 to 87.5 percent. On the deliberately anchor free interiors it correctly abstains. The stage cannot manufacture information, so its ceiling is set by identifiability, not by tuning.

A recorded negative result

Because the specification permits a gamma range of 0.4 to 2.5 and normalized cross correlation is invariant to affine intensity changes but not to gamma, tone normalization was an obvious candidate fix. Both global histogram equalization and adaptive local equalization were implemented and measured on all three datasets available at the time. Both were neutral to harmful: on the stress dataset the pass rate within 5 px fell from 60.0 percent with no tone normalization to 56.7 percent with equalization and 46.7 percent with adaptive equalization, and neither changed the primary dataset.

The explanation came from reading the reference pipeline: gamma, speckle and impulse noise are applied identically to both captures, while only shear, jitter, vignetting and radial distortion are applied asymmetrically. A monotonic tone change shared by both images is largely absorbed by a locally normalized correlation, so there was nothing for tone normalization to fix, and its cost in fine structure was real. The option remains in the configuration, switched off, with this measurement as the reason.

A recorded negative result on learning

A small convolutional re-ranker was trained to replace the statistical decision in the residual stage, using labels that are free because ground truth is known. It matched the classical rule in distribution and lost to it out of distribution, falling from 43.3 to 30.0 percent within 1 px on the held out adversarial generator. A network trained on one generator's physics did not transfer to an independent generator, and the hidden test set is by definition an independent generator. The physics grounded decision therefore remains the default and the re-ranker ships behind a flag, with the full training and comparison record under `experiments/`.

The claim this evidence supports is narrow and worth stating precisely: this re-ranker, trained on 88 degenerate pairs from one generator, did not transfer. It is not evidence that learned re-ranking cannot work here. Establishing that would require substantially more training data and domain randomisation over the generator parameters, so that the network is given a fair opportunity to fail.

The confidence regime rule, and the defect that produced it

A spot check while auditing the repository found the worst possible kind of error: a pair labelled `unique\_peak`, the most confident regime, that was wrong by 250 px. The diagnostics explained it immediately. The regime was decided by the strict equal match count alone, and that pair had one strict candidate but 4444 candidates inside the wider noise scaled tolerance. On a thoroughly degenerate surface the strict tolerance can isolate a single peak by noise, and the label then claims certainty that the evidence does not support.

The rule was re-derived from measurement rather than intuition. Across all 150 pairs from the four generators, the wide candidate pool separates correct from incorrect answers far better than the strict count: when the answer is correct the median wide pool is 1 with a 75th percentile of 2, and when it is wrong the median is 29 with a 75th percentile of 410. Thresholds were then swept:

```
| unique_peak requires | cases labelled | precision |
| --- | --- | --- |
| strict at most 1 (the old rule) | 112 | 77.7 percent |
| strict at most 1 and wide at most 1 | 67 | 98.5 percent |
| strict at most 1 and wide at most 2 | 73 | 98.6 percent |
| strict at most 1 and wide at most 4 | 81 | 91.4 percent |
| strict at most 1 and wide at most 8 | 86 | 89.5 percent |
```

A wide pool of at most 2 was selected: it is the knee of the curve, and it makes the confident label mean what its name claims. The resulting three regimes, measured over the same 150 pairs, are ``unique_peak`` at 98.6 percent precision over 73 pairs, ``residual_identified`` at 50.0 percent over 16, and ``tie_break_convention`` at 21.3 percent over 61. Accepting the first two and reacquiring on the third covers 59 percent of cases at 89.9 percent precision, against 62.0 percent if every answer is accepted blindly.

This defect is worth recording for what it says about the project's own claims. It changed no predicted coordinate and no accuracy number, so the localization result was never affected; what it corrupted was the system's account of its own certainty, which is precisely the property the abstain and reacquire policy depends on. It was found by checking a requirement rather than by any accuracy metric, because an aggregate pass rate cannot see a confidently wrong answer.

Phase 2 (Day 4-5): the magnification cliff, geometric-consistency rejection, and

Phase 2 adds a ``~20%`` absent (Set C) population and scores a ``found`` rejection decision plus a calibration AUC. Two defects surfaced and were fixed; both were found by a requirement, not by an aggregate accuracy number.

Root cause: the appearance collapse at 8×/12×

At magnifications away from the 10× nominal the reference is **decimated** rather than physically imaged, so the search template is formed by downsampling the reference and then re-upsampling. Under OpenCV's default (area-averaging) decimation the high-frequency edge content of the reference is averaged away before it is ever correlated, so at 8×/12× the template carries **less** detail than the true instance it is supposed to match. The oracle error (ground-truth window vs the correlation peak) therefore jumps from a few pixels at 10× to 350 px at 8× and 180 px at 12× — a pure appearance/cliff failure, not a search failure (the search image is fine; the **template** is degraded).

****Fix (equivalence-gated).**** The blur applied to the reference is now drawn at the **effective** zoom the matcher will actually use, ``sigma = cfg.zoom * scale / sqrt(12)``, built per scale in a blur bank (``_make_template`` / ``_band``) instead of a single approximation. At nominal 10× the new bank is bit-for-bit identical to the old path (equivalence gate, ``tests/test_template_magnification.py``: oracle `<= 5 px` across 8/9/10/11/12×, 10× unchanged within 1e-3 px). No predicted coordinate or accuracy number changed at 10×; only the previously-broken extremes were repaired.

Why raw correlation strength is anti-correlated with true presence

After the cliff fix, present pairs **still** score lower than spurious Set C matches: a degraded-but-present instance (8×/12×) produces a weak, smeared peak, while a random substrate patch can produce a sharper accidental correlation. So thresholding the raw peak (``diag["score"]``) rejects the wrong pairs. This is the reason a single threshold on correlation strength cannot reach the F1 `>= 0.90` target (it tops out at 0.877).

****Fix (geometric-consistency presence signal).**** A true instance, located anywhere in the search, aligns the full-resolution reference (warped by the predicted pose, ``zoom = cfg.zoom*scale``, over a small $\pm 3^\circ / \pm 0.15$ scale grid) with high normalized cross-correlation; a distractor does not. ``geo_consistency``

(affine `scipy` warp + `matchTemplate`) is therefore *positively* correlated with true presence, unlike the peak. It is added to `presence\_features` and used by a tiny standardized numpy logistic regression over the ten interpretable features (`peak\_score`, `num\_candidates\_wide`, `uniqueness`, `stage2\_identifiability`, `margin\_strength`, `peak\_contrast`, `peak\_contrast\_ratio`, `geo\_consistency`, `search\_noise\_sigma`, `inverted\_contrast`). The logistic model reaches **Rejection F1 = 0.9068** at threshold 0.5 on the 200-pair mixed set (156 present / 44 absent, Set C substrate-only). The earlier Day-4 weights had been fit *before* the per-scale blur fix and were mismatched to the new feature distribution; retraining on the post-T3 features is what lifted F1 from 0.892 to 0.907.

Calibration: the published `score` is $P(\text{present}) \times \text{geo_consistency}$

The rubric's calibration AUC is AUC of `score` vs *correctness* (present and within 5 px, or absent and rejected). `P(present)` alone ranks presence but is blind to localization error, so it is **anti-correlated** with that correctness label (AUC ~ 0.48). The full-resolution alignment `geo\_consistency` does rank localization correctness (AUC ~ 0.77). The published confidence is therefore the product $\text{prediction_confidence} = P(\text{present}) \times \text{geo_consistency}$, which reaches **calibration AUC 0.77** (≥ 0.75 target) while remaining in [0,1] and kept in the `score` column even when `found` = 0.

Limitations recorded, not hidden

8x/12x residual risk. The cliff is repaired to oracle ≤ 5 px, but the logistic rejection at the extremes still depends on the warp grid; the degenerate-present guard (many candidates, true instance, high geo) is preserved — those pairs stay `found` = 1.

Optical / Set D bonus = 0.0. The RGB optical modality routes through `optical\_config` and emits confident false detections 127–863 px from ground truth (20 pairs, 12 `found` = 1, 0 within 5 px). This is the bonus-only optical track and does not affect the grayscale Set A/B/C deliverable; it is documented here as a known limitation rather than blocked on.

Thermal caveat. Fanless machine throttles $\sim 1.7\times$ when hot; runtimes are compared only within the same thermal state. Cool median is 3.85 s/pair (≤ 5 s gate).

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Architecture

Design principle

The generator does not paint SEM artifacts onto a drawing. It builds a specimen (a material map and a height map in nanometer units) and then simulates two independent captures of that same specimen through a physical signal chain. Edge brightening, contrast between regions, aliasing in the zoomed out image, and the noise difference between the two images all emerge from the model instead of being applied as filters. This matters for the localization task: the search image is a genuinely different observation of the same object, exactly as on a real tool.

Coordinate frames

Specimen frame: nanometers, u rightward, v downward. The canvas covers 11.2 μm at 2 nm per pixel so that a rotated 10 μm search field plus margins always fits.

Capture frame: image pixels, x is the column, y is the row, origin at the center of the top left pixel. A capture pose is a center (u, v), a rotation theta and a pixel size. Pixel (r, c) samples the specimen at the affine image of (r + dy(r), c + dx(r)), where dx and dy are the per line scan offsets from drift, jitter and vibration.

The same two mapping functions, ``capture_to_specimen`` and ``specimen_to_capture``, are used both for rendering and for ground truth, so the recorded truth cannot disagree with the rendered images. Ground truth is the reference capture center mapped to specimen coordinates and back through the search capture transform, scan offsets included, solved with a short fixed point iteration.

Generation pipeline per pair

1. Independent random generators are spawned from the pair seed for layout, canvas physics, poses, reference capture and search capture. The two captures therefore have fully independent noise, as the problem statement requires.
2. The layout builder rasterises the chosen style onto the canvas. Structural randomisation per pair: pitches, duty cycles, phases, block sizes, line edge roughness amplitude and correlation length, contact size variation, missing contact defects, cell width sequences, contact and via presence.
3. The secondary electron canvas is computed once: material yield lookup, height map smoothing for finite sidewall slope, secant of surface tilt for edge brightening with a tilt clamp, and a directional detector asymmetry term.
4. Each capture renders through: supersampled affine sampling of the SE canvas (5x for the search capture so the point sampling to the 10 nm grid keeps its physical aliasing), anisotropic Gaussian beam blur, decimation with per line scan offsets, dielectric charging as a smooth multiplicative field over oxide regions, Poisson shot noise scaled by electron dose, Gaussian read noise, percentile tone mapping with random headroom and 8 bit quantization.
5. The search capture pose carries a random rotation (sigma 0.8 deg, clamped to 2 deg), a pixel size error up to 1.8 percent and a center offset. The reference capture carries a small residual rotation. Reference placement follows a strategy mix: uniform, deep inside a periodic array (the deliberately hard case) or near an aperiodic boundary.

Localizer pipeline

1. Impulse noise is removed by a 3 by 3 median that fires only when the outlier fraction is detectable, because salt and pepper pixels are unbounded outliers that dominate correlation sums and that Gaussian denoising spreads rather than removes.
2. The reference is blurred to a bank of plausible beam spot sizes, each band passed at the same physical cutoff as the denoised search image. Every blur additionally includes, in quadrature, the box filter equivalent of the magnification ratio, sigma equal to zoom over the square root of twelve. This matters because a search pixel integrates a zoom by zoom block of specimen, so a point sampled template would carry aliasing the search image does not have.
3. For each blur, rotation and scale hypothesis, one affine transform resamples the blurred reference onto the search pixel grid.
4. The nominal pose, no rotation and exactly the nominal magnification, is evaluated first at full resolution over the whole blur bank, because an exact decimation is by far the most likely case. If it correlates strongly the search stops there. Otherwise the wide grid over rotation to plus minus 3 degrees and magnification from 9 to 1 through 11 to 1 is screened at half resolution, the best hypotheses are rescored at full resolution and refined, and an off nominal pose is accepted only if it beats the nominal one by a margin, since a wide grid gives many chances for a wrong pose to win on noise alone. Contrast polarity is decided from the signed correlation extrema at the nominal pose.
5. On the best response map, local maxima within a tight tolerance of the global maximum form the equal match set that the centre tie break applies to, while a wider noise scaled tolerance defines the rescue pool for stage two. The tight tolerance is deliberate: the reference crop origin is sampled uniformly, so proximity to the image centre carries no information about correctness, and applying the tie break to candidates that score measurably worse can only lose accuracy. A single equal match is returned directly with parabolic sub pixel refinement.
5. When the surface is degenerate, the residual disambiguation stage runs: cell to cell reference subtraction, the standard array inspection technique, applied to localization. The pixelwise median over the best aligned candidate windows estimates the shared periodic content. The template's deviation field is what remains after projecting out the median and its sub pixel shift terms (a least squares fit of the median and its two gradient images), which removes the systematic lattice leakage that would otherwise correlate with every aligned window. The normalized correlation between this deviation field and every window's deviation field is then computed densely in closed form (two raw correlations plus integral image sums), so every candidate position is scored with no selection

bias regardless of how many near ties exist. The decision is a robust z score of the best candidate against the candidate population, which self calibrates to pool size and noise; small pools use a fixed margin instead. A decisive winner is returned because the deviation field is the only physically identifying information a periodic array offers; otherwise the problem statement tie break applies among the strict equal match set: the candidate closest to the search image center.

The diagnostics expose which regime produced each answer (single candidate, decisive residual, or tie break convention) plus the residual z score, so every output carries its own confidence.

An optional learned decision can replace the z score rule (see the re-ranker section of the README): a small CNN scores each candidate from the same evidence the statistical rule uses, the search window, template and both deviation fields plus the two scalar scores, with a learnable abstain class trained on pools whose true site is absent. It changes only the decision among already found candidates, never the candidate search, so its worst case is bounded by the classical pipeline's candidate set, and abstention still falls through to the mandated tie break.

Failure modes by construction

The generator can place the reference deep inside a uniform array (placement ``deep_array``). There the correlation surface has a lattice of near equal peaks and the tie break rule decides, which is exactly the failure regime the problem statement asks to be demonstrated and explained. The evaluation harness separates metrics by placement so this regime is measured rather than hidden.

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Citations

Every augmentation choice, noise model and structural parameter in the generator, and the core algorithmic choices of the localizer, mapped to public literature. Reference list at the end; bracketed numbers link each choice to its sources.

SEM image formation

Edge brightening via the secant law. Secondary electron yield rises with the local tilt of the surface approximately as $1/\cos$ of the tilt angle, because inclined surfaces give escaping secondaries a larger interaction volume near the surface. We therefore compute the height map gradient, convert it to a tilt angle, and multiply the emitted signal by $\sec$ of that angle with a clamp near grazing incidence. Bright feature edges then emerge physically instead of being painted with an edge filter. [1] [2] [3]

Material dependent yields. Secondary electron yield differs by material at fixed beam energy; oxides yield notably more than clean silicon, metals sit in between depending on energy. We assign relative yields (silicon 1.0, silicon dioxide 1.45, gate metal 1.2, tungsten 1.5, nitride 1.3) with a per pair random jitter of 10 percent, giving the material contrast visible in real SEM images while acknowledging the published spread. [1] [3]

Beam point spread function. The finite probe is modelled as an anisotropic Gaussian whose sigma is drawn from 1.5 to 2.6 nm for the high magnification capture and 4 to 12 nm for the wide field capture, with up to 35 percent astigmatic ellipticity, following the Gaussian probe treatment used in SEM image simulation for resolution metrology. [2] [4]

Poisson shot noise scaled by dose. SEM noise is dominated by counting statistics of the primary and secondary electrons, so pixel noise is Poisson in the collected signal with a Gaussian detector and amplifier contribution on top. We draw an electron dose per pixel (700 to 2000 electrons for the reference, 80 to 300 for the search capture), sample Poisson counts of signal times dose, and add Gaussian read noise of 2 to 8 electrons. The search image is therefore always the noisier capture, as the problem statement requires of the test data, and the noise fields of the two captures come from independent generators because they are separate physical scans. [5] [1] [2]

Dielectric charging. Insulating regions accumulate charge under the beam and shift their apparent

brightness on the scale of the scan, so we multiply oxide regions by a smooth random field of 3 to 9 percent amplitude with 120 to 320 nm correlation length. [6] [2]

Scan drift, line jitter and vibration. Stage and column drift during frame acquisition displaces successive scan lines, and environmental vibration adds a periodic component; artificial SEM image generators include exactly these distortions. We add a smooth drift trajectory (up to 3 reference pixels, up to 1.2 search pixels over the frame), an autoregressive per line jitter and a sinusoidal vibration term, applied as per line sampling offsets before the noise stage, and the recorded ground truth accounts for them. This is also thematically the point of the exercise, since stage drift is the reason navigation error recovery exists. [4] [7] [8]

Aliasing preserved at the wide field of view. At 10 nm per pixel, fin and line pitches of 26 to 60 nm sit near the sampling limit. The wide capture is rendered on a 2 nm supersampled grid, blurred by the beam, then point sampled to the 10 nm grid, so the aliasing a real point sampling scan produces is preserved instead of suppressed by area averaging. [2] [8]

Tone mapping. Operators set brightness and contrast per capture, so each image is percentile normalized with random headroom before 8 bit quantization, which decorrelates the gray scales of the two captures. [1]

DRAM structural parameters

Cell geometry. The layout follows the 6F2 buried word line DRAM cell: word line pitch 2F, bit line pitch 3F, with F drawn from 16 to 22 nm to span reported product nodes. Storage node contacts sit between bit lines at half pitch offsets with 4 percent size variation and a small missing contact probability as defects. [9] [10] [11]

Array organisation. DRAM arrays are multi divided into mats bounded by bit line sense amplifier stripes and local word line driver stripes, with local lines typically 256 to 512 cells long; shorter local lines appear in speed optimised designs. We draw mat sizes of 4.5 to 9 μm , corresponding to roughly 128 to 250 cells per local line at the simulated feature sizes, with stripe widths of 300 to 560 nm filled with quasi random periphery blocks. These stripes are the aperiodic anchors that make localization inside an otherwise repeating array possible at all. [10] [11] [23]

Defect density. Storage node contacts carry 4 percent size variation and a small missing contact probability. The upper part of the probability range reflects that navigation error recovery by definition happens at sites an inspection tool chose to revisit, which are disproportionately sites with measurable anomalies. [9] [11]

FinFET structural parameters

Fin and gate grids. Fin pitch is drawn from 26 to 36 nm and contacted gate pitch from 50 to 60 nm, spanning reported 10 nm and 7 nm class technologies; fin width is 30 to 40 percent of pitch and fin height 46 nm within the reported 40 to 55 nm range. [12] [13] [14]

Standard cell structure. Logic is organised in rows of 6 to 9 fins with cells of 2 to 9 gate pitches, diffusion breaks of 0.8 gate pitch at cell boundaries, trench contacts between gates with 60 percent occupancy and sparse vias, following published standard cell construction for FinFET nodes. One rectangular SRAM block is rendered perfectly regular to provide the highly periodic hard region the test set is stated to contain. [14] [12]

Line edge roughness. Fin, gate, word line and bit line edges carry correlated roughness with sigma 1 to 2.4 nm and correlation length 15 to 40 nm, matching reported LER magnitudes and correlation lengths for litho and etch defined lines. [15] [16] [17]

Optical modality (bonus)

Diffraction limited resolution. The brightfield point spread is modelled as a Gaussian of sigma 0.21 λ over NA per channel, the standard Gaussian approximation of the Airy pattern, with chromatic variation following wavelength. At visible wavelengths and NA near 0.9 this is roughly 130 nm, so nanometer scale array features are unresolved and only mesoscale structure (mats, stripes, blocks)

carries localization information, which matches the production division of labor between optical coarse alignment and electron beam fine addressing. [24] [25] [27]

Thin film interference color. Oxide regions are colored by two beam interference with phase $4\pi n t$ over λ for film thickness t and index 1.46, which is why wafer dielectric films show thickness dependent colors under brightfield inspection. [24] [26]

Photon noise and camera effects. Per channel Poisson shot noise scaled by exposure (the wide field capture receives far fewer photons per pixel), small Gaussian read noise, radial vignetting, illumination tilt and white balance jitter model the camera chain; optical tools are full field cameras, so no scan line artifacts apply. [25] [26]

Localizer choices

Normalized cross correlation. NCC is the standard robust similarity for template localization under linear intensity changes, computed with the FFT accelerated formulation. [18]

Rotation and scale search. The relative pose between captures is handled by a hypothesis grid over rotation and scale with coarse to fine refinement, the discretised counterpart of FFT based rotation and scale registration. [19] [20]

Why not sparse features. Keypoint descriptors such as SIFT are ambiguous on repeating structures because hundreds of near identical keypoints exist per frame, which is the documented failure regime for periodic patterns; dense correlation with a tie break rule is the appropriate tool. [21]

Matched formation template. Blurring the reference to the search optics resolution and point sampling it onto the search grid reproduces the degradation chain of the search image, following the matched filter principle that correlation is optimal when the template matches the observed signal formation. [18] [2]

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All entries verified against the publisher or an authoritative index.

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